

WASP-52b

R-1465

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_151001 · created 2026-08-01T11:15:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457296.7408 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.05 +/- 0.059
Transit depth (Rp/Rs)^2	0.25 +/- 0.59 [%]
Orbital Inclination (inc)	80.56 +/- 2.88
Airmass coefficient 1 (a1)	1.02 +/- 0.22
Airmass coefficient 2 (a2)	0.0 +/- 0.74
Scatter in the residuals of the lightcurve fit is	21.41 %
Best Comparison Star	#1 - [343, 293]
Optimal Method	PSF photometry
Transit Duration (day)	0.015 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.05 ± 0.059 vs 0.1646 (deviation 1.1σ)
Duration fit vs published	0.015 d vs 0.0758 d (deviation 1.44σ)
Mid-time O-C	-4.3 min
Depth SNR	0.5
Residual scatter	21.41 %

Light curve

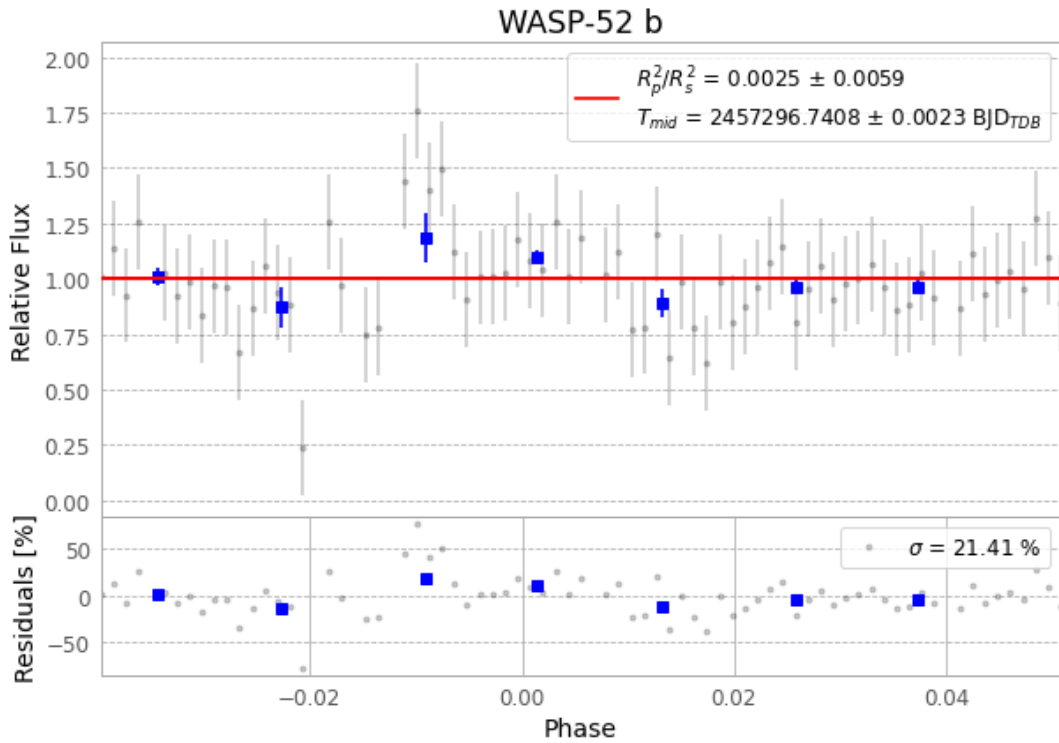


Figure 1 — FinalLightCurve_WASP-52 b_2015-09-30.png (SHA-256 18dc4a0f4b82baee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [613, 105], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 571217e10512b20f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52151001035712.FITS → WASP-52151001074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-151001.log (a0f6f350aacf...), FinalLightCurve_WASP-52 b_2015-09-30.png (18dc4a0f4b82...), FinalLightCurve_WASP-52 b_2015-09-30.csv (a1ce28ecc233...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 11:15Z.